

Idealne tekočine

$$\rho \frac{d\vec{v}}{dt} = \rho \left[\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla) \vec{v} \right] = -\nabla p + \vec{f} \quad (\text{Euler})$$

$\vec{f}$ volumska gost. sila

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0 \quad (\text{kont. en.})$$

$$\frac{\partial \vec{v}}{\partial t} = \vec{v} \lambda (\nabla \times \vec{v}) - \nabla \left(\frac{1}{2} v^2 + \frac{\nabla p}{\rho} \right)$$

Brez vrtinčen tok: $\nabla \times \vec{v} = 0$

Nestisljiv tok: $\nabla \cdot \vec{v} = 0$

$$\nabla \times (\vec{a} \times \vec{b}) = (\nabla \cdot \vec{b}) \vec{a} - (\vec{a} \cdot \nabla) \vec{b} +$$

$$+ \vec{b} \nabla (\nabla \cdot \vec{a}) - \vec{a} (\nabla \cdot \vec{b})$$

$\vec{\omega} = \nabla \times \vec{v}$ vrtinčnost

$$\frac{\partial \vec{\omega}}{\partial t} + (\vec{v} \cdot \nabla) \vec{\omega} = (\vec{\omega} \cdot \nabla) \vec{v} \quad (\text{Helmholtz})$$

$$\nabla \left(\frac{1}{2} v^2 + \frac{\nabla p}{\rho} + \frac{\partial \psi}{\partial t} \right) = 0 \quad (\text{Bernoulli})$$

$$\vec{v} = -\nabla \psi$$

$$\vec{v}(\vec{r}) = \frac{1}{4\pi} \int d\vec{r}' \frac{\vec{\omega}(\vec{r}') \times (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3} =$$

$$= \frac{\Gamma}{4\pi} \int \frac{d\vec{l} \times (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3} \quad (\text{Biot-Savart})$$

$$\Gamma = \oint \vec{v} \cdot d\vec{l} \quad \text{cirkulacija}$$

Viskozne tekočine

$$\rho \frac{d\vec{v}}{dt} = -\nabla p + \eta \nabla^2 \vec{v} + \left(\nabla \cdot \frac{1}{3} \eta \right) \nabla \vec{v} \cdot \vec{f}$$

(Navier-Stokes)

$$\tau_v = \frac{\eta l^2}{\eta} \quad (\text{čas za stac. polje})$$

$$Re = \frac{\rho (\vec{v} \cdot \nabla) \vec{v}}{\eta \nabla^2 \vec{v}} = \frac{\rho v l}{\eta}$$

$$St = \frac{l}{\tau_v} = \frac{|\frac{\partial \vec{v}}{\partial t}| \cdot |(\vec{v} \cdot \nabla) \vec{v}|}{\eta \nabla^2 \vec{v}} = \frac{1}{Re}$$

$$\omega \tau = \tau_v$$

nestisljiv stac. tok + Re majhen: $0 = \nabla p + \eta \nabla^2 \vec{v} + \vec{f}$

$\vec{f} = 0, \rho = \text{konst.}$ vrtinčnost $\frac{\partial \vec{\omega}}{\partial t} + (\vec{v} \cdot \nabla) \vec{\omega} = (\vec{\omega} \cdot \nabla) \vec{v} + \frac{\eta}{\rho} \nabla^2 \vec{\omega}$

Sila na tekočino: $2\eta \int (\nabla v_{ij}) dV = 2\eta \int v_{ij} v_{ij} dV = \rho = -\omega M(\text{moč})$

$$\frac{d\vec{F}}{ds} = \sigma_{ij} n_j, \quad \sigma_{ij} = 2\eta v_{ij}, \quad v_{ij} = \frac{1}{2} (\partial_j v_i + \partial_i v_j), \quad dM = R dF$$

2D tekočina: ($\nabla p = 0$)

Tokovna funkcija ψ :

$$v_x = \frac{\partial \psi}{\partial y}, \quad v_y = -\frac{\partial \psi}{\partial x} = \frac{\partial \phi}{\partial y}$$

(bezvrtinčen, nestisljiv tok)

$$\text{pretok } Q_{\eta\eta} = \psi(\vec{r}_2) - \psi(\vec{r}_1)$$

$$w = \phi + i\psi, \quad z = x + iy \Rightarrow dw/dz = v_x - iv_y$$

$$\vec{g} = g\vec{v} \text{ gost. gib. kol.} \Rightarrow \frac{\partial g_i}{\partial t} + \frac{\partial \pi_{ij}}{\partial x_j} = f_i$$

$$\pi_{ij} = \rho v_i v_j + p \delta_{ij} \quad (\text{Euler})$$

$$\int \frac{\partial \pi_{ij}}{\partial x_j} dV = \int \pi_{ij} ds_j$$

Kelvinov izrek: vrtinčnost ni se prenika s hitrostjo tekočine na mestu niti

2D tekočina:

$$\psi = y v_x, \quad \phi = x v_x \Rightarrow w = z v_x$$

poiščeno preslikavo v željeno območje

$$\frac{z+i}{iz+1}$$

$$\frac{z-i}{-z-i}$$

$$\frac{z-2i}{-iz}$$

$$\frac{z-i}{z+i} e^{i\pi/4}$$

$$z + \frac{A}{z}$$

$$r = b \Rightarrow \ln(b/a)$$

Bessel

$$x^2 \frac{d^2 z}{dx^2} + x \frac{dz}{dx} + (x^2 - \alpha^2)z = 0 \rightarrow J_\alpha, N_\alpha$$

$$\frac{d^2 z}{dx^2} + \frac{2}{x} \frac{dz}{dx} + \left(1 - \frac{(1+\alpha^2)}{x^2}\right)z = 0 \rightarrow j_\alpha, n_\alpha \text{ (sferni Bessel)}$$

Nabla u v sfernih koord.

$$\nabla f = \frac{\partial f}{\partial r} \hat{e}_r + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{e}_\theta + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \varphi} \hat{e}_\varphi$$

$$\nabla^2 f = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \varphi^2}$$

$$\nabla \cdot \vec{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta A_\theta) + \frac{1}{r \sin \theta} \frac{\partial A_\varphi}{\partial \varphi}$$

$$\nabla_x \vec{A} = \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (A_\varphi \sin \theta) - \frac{\partial A_\theta}{\partial \varphi} \right] \hat{e}_r + \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial A_r}{\partial \varphi} - \frac{\partial}{\partial r} (r A_\varphi) \right] \hat{e}_\theta + \frac{1}{r} \left[\frac{\partial}{\partial r} (r A_\theta) - \frac{\partial A_r}{\partial \theta} \right] \hat{e}_\varphi$$

Nabla u v kart. koord.

$$\nabla f = \frac{\partial f}{\partial x} \hat{e}_x + \frac{\partial f}{\partial y} \hat{e}_y + \frac{\partial f}{\partial z} \hat{e}_z$$

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

$$\nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$\nabla \wedge \vec{A} = \left[\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right] \hat{e}_x + \left[\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right] \hat{e}_y + \left[\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right] \hat{e}_z$$

$$\nabla(fg) = f \nabla g + g \nabla f$$

$$\nabla(\vec{u} \cdot \vec{v}) = \vec{u} \times (\nabla \times \vec{v}) + \vec{v} \times (\nabla \times \vec{u}) + (\vec{u} \cdot \nabla) \vec{v} + (\vec{v} \cdot \nabla) \vec{u}$$

Nabla u v cil. koord.

$$\nabla f = \frac{\partial f}{\partial \rho} \hat{e}_\rho + \frac{1}{\rho} \frac{\partial f}{\partial \varphi} \hat{e}_\varphi + \frac{\partial f}{\partial z} \hat{e}_z$$

$$\nabla^2 f = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial f}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 f}{\partial \varphi^2} + \frac{\partial^2 f}{\partial z^2}$$

$$\nabla \cdot \vec{A} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho A_\rho) + \frac{1}{\rho} \frac{\partial A_\varphi}{\partial \varphi} + \frac{\partial A_z}{\partial z}$$

$$\nabla \wedge \vec{A} = \left[\frac{1}{\rho} \frac{\partial A_z}{\partial \varphi} - \frac{\partial A_\varphi}{\partial z} \right] \hat{e}_\rho + \left[\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho} \right] \hat{e}_\varphi + \frac{1}{\rho} \left[\frac{\partial}{\partial \rho} (\rho A_\varphi) - \frac{\partial A_\rho}{\partial \varphi} \right] \hat{e}_z$$

$$\nabla \cdot (f \vec{u}) = f \nabla \cdot \vec{u} + \vec{u} \cdot \nabla f$$

$$\nabla \cdot (\vec{u} \times \vec{v}) = \vec{v} \cdot (\nabla \times \vec{u}) - \vec{u} \cdot (\nabla \times \vec{v})$$

$$\nabla \times (\nabla f) = 0, \nabla \cdot (\nabla \times \vec{f}) = 0$$

$$\nabla \wedge (f \vec{v}) = (\nabla f) \times \vec{v} + f (\nabla \times \vec{v})$$

$$\nabla \wedge (\vec{u} \times \vec{v}) = \vec{u} (\nabla \cdot \vec{v}) - \vec{v} (\nabla \cdot \vec{u}) + (\vec{v} \cdot \nabla) \vec{u} - (\vec{u} \cdot \nabla) \vec{v}$$

$$\nabla(\vec{v}^2/2) = \vec{v} \times (\nabla \times \vec{v}) + (\vec{v} \cdot \nabla) \vec{v}$$

$$\nabla \nabla \cdot = \nabla^2 + \nabla \times \nabla \times$$

Greenove funkcije:

Vij v cilindričnih

$$V_{rr} = \partial_r V_r \quad V_{\varphi\varphi} = \frac{1}{r} \partial_\varphi V_\varphi + \frac{1}{r} V_r$$

$$V_{zz} = \partial_z V_z \quad V_{rz} = \frac{1}{2} (\partial_z V_r + \partial_r V_z)$$

$$V_{r\varphi} = \frac{1}{2} \left(\partial_r V_\varphi - \frac{1}{r} V_\varphi + \frac{1}{r} \partial_\varphi V_r \right)$$

$$V_{\varphi z} = \frac{1}{2} \left(\frac{1}{r} \partial_\varphi V_z + \partial_z V_\varphi \right)$$

Poisson

$$\nabla^2 \phi = -\rho(\vec{r})$$

$$\phi = \int \frac{\rho(\vec{r}')}{4\pi |\vec{r} - \vec{r}'|} d^3r'$$

$$\frac{\partial u}{\partial t} - D \frac{\partial^2 u}{\partial z^2} = \delta(z) \delta(t), z \in \mathbb{R}$$

$$u(z,t) = \frac{1}{\sqrt{4\pi Dt}} e^{-z^2/4Dt}$$

več dimenzij
→ parnož. reš.
med sabo

Ena plošča pri miru,
druga sinusno nihaj $\frac{eV}{m}$

• vdorna globina $\sqrt{\frac{2\eta}{\rho \omega}}$

• fazni zamik $\Delta\varphi = \sqrt{\frac{\rho \omega}{2\eta}} z$

Sila na veliki okrogli plošči (r=a),
ki jo razmikamo s hitr. u, vnes.
nest. tekočin

$$F = -\frac{3\pi}{2} \frac{\eta a^4}{h^3} u$$

Vij v sfernih

$$V_{rr} = \partial_r V_r \quad V_{\theta\theta} = \frac{1}{r} \partial_\theta V_\theta + \frac{1}{r} V_r$$

$$V_{\varphi\varphi} = \frac{1}{r \sin \theta} \partial_\varphi V_\varphi + \frac{1}{r} V_r + \frac{1}{r \tan \theta} V_\theta$$

$$V_{r\theta} = \frac{1}{2} \left(\partial_r V_\theta - \frac{1}{r} V_\theta + \frac{1}{r} \partial_\theta V_r \right)$$

$$V_{r\varphi} = \frac{1}{2} \left(\partial_r V_\varphi - \frac{1}{r} V_\varphi + \frac{1}{r \sin \theta} \partial_\varphi V_r \right)$$

$$V_{\theta\varphi} = \frac{1}{2} \left(\frac{1}{r} \partial_\theta V_\varphi - \frac{1}{r \tan \theta} V_\varphi + \frac{1}{r \sin \theta} \partial_\varphi V_\theta \right)$$